

FIN ST1362–ST1451

Interval-Certified OA Optimization,
Composite Identifiability and Sequential Inference

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Six research rounds — 90 programmes
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This report upgrades two strong numerical results from Release 10.98 to outward interval certificates for a frozen decimal spectral model: the unique return-gap maximizer and positive four-time separation from a declared dephased-quantum nuisance box. The result remains conditional on the frozen Operational/State package and is not laboratory evidence.

Abstract

ST1362–ST1451 strengthen the frozen Operational/State discrimination programme of Release 10.98. A decimal spectral model is declared exact, with seven eigenvalues and multiplicities $(1, 2, 2, 2, 2, 2, 1)$. Outward interval arithmetic then proves that the absolute classical–quantum return gap on $[0, 10]$ has one global maximum. Its time lies in

$$[0.59453078, 0.59453079]$$

and its value lies in

$$[0.4112409998076315, 0.41124099980763185].$$

The proof covers $[0, 10] \setminus [0.59, 0.60]$ with 4995 Taylor interval boxes, proves positivity and strict concavity on the remaining interval, and brackets the unique derivative zero. This upgrades the previous dense-grid result to a theorem for the frozen decimal spectrum, not for symbolic kernel parameters.

The four-time composite problem is also certified. A complete cover of 1000 rectangles in $(\alpha, \gamma) \in [0.9, 1.1] \times [0, 100]$ proves that no member of the declared energy-dephased quantum family reproduces the classical return curve at times $(0.3, 0.6, 1.2, 2.0)$. The residual sum of squares satisfies

$$\text{RSS} \geq 0.046692011213846946,$$

so at least one time has absolute residual at least 0.10804167160619894. A point candidate has $\text{RSS} 0.05239469837327923$; its optimizer status is not proved.

Preparation, detector and loss budgets are made explicit. The full twelve-vertex likelihood reduces without information loss to seven cyclic distance classes. Its Chernoff information is 0.13647980004881377, versus 0.10028599984500335 for binary return. A conservative Chernoff bound gives 29 full-record shots for equal-prior error below one percent in the two simple hypotheses. The interval composite certificate yields a much more conservative 1146 binary shots per time by a Hoeffding union bound.

For the simple seven-class models, the likelihood ratio is an anytime-valid e-process. Symmetric boundaries 100 and 0.01 control either wrong decision by one percent under optional stopping. A naively refitted composite likelihood is not automatically an e-process. Protocol 10.99, its fail-closed sequential validator and synthetic fixtures implement these distinctions.

No state ontology, dimensional clock value, apparatus, raw natural event, independent custody or physical validation is supplied. The certificates make a conditional *OA* mathematically sharper; they do not promote it into strict FIN physics.

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1 Frozen decimal spectral model

The interval theorems treat the following decimal inputs as exact:

$$\lambda = (0, 0.75412115420707981, 1.5770495144276093, \\ 1.9614068619764458, 2.1995688493332102, \\ 2.2986062720790965, 2.3421820411462999),$$

with multiplicities $(1, 2, 2, 2, 2, 1)$. This freezes a reproducible finite problem and removes binary floating ambiguity inside the certificate.

It does not certify an exact symbolic spectrum from $(\omega, \phi, \beta, \eta)$ or an interval matrix containing every reconstruction error. That lift is a recommended next problem.

Define

$$C(t) = \frac{1}{12} \sum_k m_k e^{-\lambda_k t}, \quad qquad a(t) = \frac{1}{12} \sum_k m_k e^{-i\lambda_k t}, \quad qquad Q(t) = |a(t)|^2,$$

and $F(t) = Q(t) - C(t)$. The optimized quantity is $|F(t)|$.

2 Round I: ST1362–ST1376 — certified time optimum

2.1 Global cover

The region

$$[0, 10] \setminus [0.59, 0.60]$$

is partitioned into 4995 intervals of width 0.002. For a box I with midpoint m and radius $r = 0.001$, the mean-value enclosure is

$$|F(I)| \leq |F(m)| + r \sup_{t \in I} |F'(t)|,$$

with every function evaluation outward-rounded by interval arithmetic.

At $t = 0.5945308$ the certified lower value is 0.4112409998076313. The largest upper bound outside the target interval is 0.41123791023521883, attained by the cover cell $[0.588, 0.59]$. The strict exclusion margin is

$$3.089572412462438 \times 10^{-6} > 0.$$

2.2 Unique maximum inside the target interval

On $[0.59, 0.60]$, interval arithmetic gives

$$F(t) \in [0.3966679604508919, 0.425775515327064],$$

so $|F| = F$, and

$$F''(t) \in [-1.6864459900506021, -1.3685959622948138].$$

Thus F is strictly concave. The endpoint derivative enclosures are

$$F'(0.59453078) \in [2.5452303868190474, 2.5452303868190486] \times 10^{-9},$$

$$F'(0.59453079) = -1.2746103229835985 \times 10^{-8}.$$

The derivative has one and only one zero in the bracket.

Theorem 1 (unique frozen-decimal return-gap maximum). *For the declared decimal spectrum, $|Q(t) - C(t)|$ has exactly one global maximizer on $[0, 10]$. It lies in $[0.59453078, 0.59453079]$, and the maximum value lies in*

$$[0.4112409998076315, 0.41124099980763185].$$

This converts a previous strong numerical observation into an interval theorem within the declared input scope.

3 Round II: ST1377–ST1391 — composite identifiability

The nuisance family is the energy-dephased quantum return

$$Q_{\alpha,\gamma}(t) = \frac{1}{12^2} \sum_{j,k} m_j m_k e^{-\frac{1}{2}\gamma(\lambda_j - \lambda_k)^2 \alpha t} \cos((\lambda_j - \lambda_k) \alpha t),$$

with

$$0.9 \leq \alpha \leq 1.1, 0 \leq \gamma \leq 100.$$

The classical target is sampled at $T = (0.3, 0.6, 1.2, 2.0)$, and

$$R(\alpha, \gamma) = \sum_{t \in T} (Q_{\alpha,\gamma}(t) - C(t))^2.$$

The nuisance rectangle is covered by 20 clock cells of width 0.01 and 50 dephasing cells of width 2, for 1000 outward interval cells. Summing the lower bounds of the four squared residual intervals gives

$$R(\alpha, \gamma) \geq 0.046692011213846946.$$

The weakest cover cell is $\alpha \in [1.09, 1.10]$, $\gamma \in [10, 12]$; this need not contain the actual optimizer. The point $(1.1, 12.373117825382856)$ gives

$$R = 0.05239469837327923.$$

Theorem 2 (four-time composite identifiability). *No quantum model in the declared (α, γ) box exactly reproduces the four classical return probabilities. At least one of the four absolute residuals is at least*

$$\sqrt{0.046692011213846946/4} = 0.10804167160619894.$$

This is not a theorem over all Lindblad dynamics. The clock band and the energy-dephasing family are paid premises.

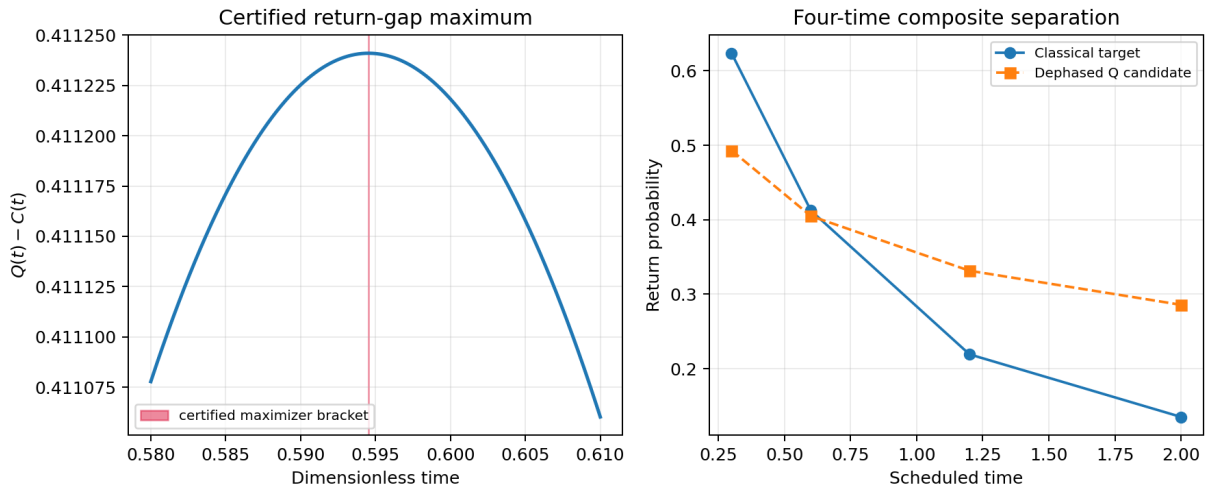


Figure 1: Left: the certified maximizer bracket (visually thinner than a pixel). Right: the four-time classical target and one near-minimizing dephased quantum candidate.

4 Round III: ST1392–ST1406 — calibration and errors

At the practical time $t = 0.6$, the ideal return gap is

$$D = 0.41121822546536.$$

If both hypotheses suffer independent unknown scalar probability errors of at most ϵ , the triangle inequality gives

$$D_{\text{observed}} \geq D - 2\epsilon.$$

Thus binary-return identifiability is guaranteed for $\epsilon < D/2 \approx 0.2056091$. Assigning independent preparation and detector budgets 0.05 each yields the conservative combined bound

$$D_{\text{observed}} \geq D - 2(0.05 + 0.05) = 0.21121822546536 > 0.$$

For common uniform source contamination,

$$p'_H = (1 - \epsilon)p_H + \epsilon/12,$$

the gap scales exactly by $1 - \epsilon$. Ten percent contamination therefore retains a gap about 0.3701.

A binary confusion channel with false-positive rate a and false-negative rate b maps

$$p \mapsto a + (1 - a - b)p,$$

and scales a common calibrated gap by $|1 - a - b|$. The surface $a + b = 1$ is nonidentifying.

Known uniform loss preserves click-conditioned vertex probabilities but changes the attempt-level click rate. If no-clicks are discarded, unknown loss is invisible. Vertex-dependent efficiencies require a calibrated confusion/efficiency matrix and are not covered by the scalar model.

5 Round IV: ST1407–ST1421 — full likelihood

At $t = 0.6$, the full event log-likelihood ratio is

$$\ell(x) = \log \frac{q_x}{p_x}, \text{quad} L_n = \sum_x n_x \ell(x).$$

Reflection symmetry makes the cyclic distance $d = 0, 1, \dots, 6$ a sufficient seven-class statistic: the vertices $+d$ and $-d$ have identical likelihood ratios. Thus the twelve labels compress to seven without information loss for the frozen simple hypotheses.

The full directional divergences are

$$D(C\|Q) = 0.706146122009315, \text{quad} D(Q\|C) = 0.4134894066196461.$$

The optimized Chernoff information is

$$\mathcal{C}_{\text{full}} = 0.13647980004881377,$$

compared with $\mathcal{C}_{\text{return}} = 0.10028599984500335$. The gain ratio is about 1.36. The single-event total variation is 0.4112182254653598, giving equal-prior Bayes error 0.2943908872673201.

The standard product Chernoff bound

$$P_e(n) \leq \frac{1}{2} e^{-nC}$$

gives 29 full-record shots as a sufficient, not minimal, bound for error below one percent. The analogous binary Chernoff bound needs 40; direct exact binomial tails from Release 10.98 happened to need only 29.

For the composite certificate, the fact that at least one time differs by $\delta = 0.1080416716$ gives a conservative Hoeffding/union design of 1146 binary shots per time, or 4584 total. This is proof-safe but not optimized.

6 Round V: ST1422–ST1436 — anytime-valid inference

For iid seven-class events define

$$E_n = \prod_{i=1}^n \frac{q(X_i)}{p(X_i)}.$$

Under the classical simple hypothesis, (E_n) is a nonnegative mean-one martingale. Under the quantum simple hypothesis, $(1/E_n)$ is one. Ville’s inequality therefore proves that the symmetric rule

$$E_n \geq 100 \Rightarrow Q, qquad E_n \leq 0.01 \Rightarrow C$$

has either wrong-boundary probability at most 0.01, even under optional stopping.

Expected log-likelihood growth is 0.4134894 per event under Q and -0.7061461 under C . Dividing log 100 by these rates gives rough Wald scales of 11.14 and 6.52 events, respectively; overshoot makes these approximations, not guarantees.

Optional inspection is safe only because the model and likelihood are frozen. Maximizing a nuisance likelihood after each event does not automatically produce an e-process. A valid composite sequential test requires a predeclared mixture, test martingale or another proved domination construction.

Protocol `FIN_ST1422_ST1436_Protocol_10_99.json` freezes the decimal spectrum, seven-class probabilities, sequential boundaries, composite box, certified RSS lower bound, calibration budgets and role separation. The fail-closed validator rejects invalid classes and malformed protocols. Seeded synthetic fixtures stop on the correct boundaries at 6 and 9 events; they are code tests, not data.

7 Round VI: ST1437–ST1451 — final status

Result	Status	Boundary
unique return-gap maximizer	[Interval-proven]	frozen decimal spectrum, $[0, 10]$
four-time nuisance separation	[Interval-proven]	$\alpha \in [.9, 1.1]$, energy dephasing $[0, 100]$
preparation/detector budgets	[Proven bounds]	scalar/common models
seven-class sufficient likelihood	[Proven]	frozen simple hypotheses
29-shot Chernoff sufficiency	[Proven bound]	full simple-record iid model
1146 shots/time composite design	[Proven conservative]	Hoeffding union bound
anytime-valid LR boundaries	[Proven]	simple hypotheses only
composite e-process	[Open]	naive plug-in invalid
protocol/validator	[Constructed/tested]	synthetic fixtures
laboratory realization	[Absent]	no apparatus/events/custody

These results strengthen the conditional OA . They do not derive OA from the strict kernel and do not choose a physical state ontology.

The protocol distinguishes mass-mixing dynamics from coherent unitary dynamics inside the proposed total state. It does not observe literal annihilation or nonexistence of the whole nadsoliton.

8 Recommended next programmes

1. lift decimal certificates to interval uncertainty in the entries of A ;
2. certify eigenvalue clusters from an interval matrix;
3. solve the minimax four-time likelihood allocation;
4. construct a valid mixture e-process over (α, γ) ;

5. add a calibrated preparation simplex;
6. add a detector confusion/efficiency polytope;
7. optimize time allocation rather than equal counts;
8. use seven-class sequential records in the composite test;
9. transport the protocol through the exact $12 \rightarrow 24$ refinement;
10. derive one platform-specific forward model without claiming realization;
11. freeze independent calibration and custody forms;
12. seek external review before any event collection.

Stop after a spectrum outside the frozen model, nuisance outside the certified box, unregistered detector law, postselection without no-clicks, adaptive plug-in likelihood or post-unblinding protocol edit.

9 Final conclusion

The main numerical discoveries of Release 10.98 are now theorem-grade inside a precisely frozen decimal model. The optimal return time is unique, and a calibrated four-time record cannot be exactly reproduced by the declared dephased-quantum family. Full spatial records and anytime-valid likelihood ratios materially improve the simple-hypothesis test. The remaining frontier is no longer whether a separation exists, but whether it survives interval uncertainty in the operator, a complete hardware nuisance model and independent physical realization.

A Six-round ledger

Range	Object	Verdict
ST1362–ST1376	interval time optimization	unique global maximum certified
ST1377–ST1391	composite interval cover	positive RSS lower bound
ST1392–ST1406	preparation/detector/loss	explicit calibration budgets
ST1407–ST1421	full multinomial likelihood	seven sufficient classes and shot bounds
ST1422–ST1436	sequential inference	simple-model e-process and protocol 10.99
ST1437–ST1451	synthesis	conditional theorem upgrade, evidence absent

B Reproducibility

The authoritative result sets run from `FIN_ST1362_ST1376_Results.json` through `FIN_ST1437_ST1451_Results.json`. Each contains fifteen programmes and packet SHA-256 references. The combined test `test_fin_st1362_st1451.py` verifies all ninety packets, interval margins, composite lower bound, error budgets, likelihood gain, sequential protocol, figure and nonpromotion gates.

C Nonclaims

No symbolic-kernel interval theorem, complete open-quantum library, physical clock calibration, state ontology, apparatus, raw event, independent custody, legacy bridge, selector, Standard Model, gravity, L_{total} or Theory of Everything is established. All future FIN PDFs remain English.

References

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